

MICROBIOLOGY

CHAPTER 7

MICROBIAL GROWTH



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Reference:
Slide and
Dr sereen notes

Chapter 7: Microbial Growth

We're going to learn about: \ (^o^) ^ (^-^) /

1. Reproductive strategies and bacterial cell cycles
2. Environmental factors that affect microbial growth
3. Microbial Growth in natural environments and in the lab
4. Growth curves
5. Measurement of microbial growth and direct measurement of cell numbers
6. Chemostats and Turbidostats (Continuous culture)

🌸 Reproductive strategies and bacterial cell cycles

Reproductive strategies

- The reproductive strategies of eukaryotic microbes
 - asexual and sexual, haploid or diploid

العukaryotic microbes تختلف بشكل كبير عن البكتيريا فيما يخص انقسام الخلية؛ كثير منهم بقدروا ينقسموا asexually عن طريق mitosis وبنفس الوقت بقدروا ينقسموا sexually عن طريق meiosis بالإضافة الى انهم ممكن يكونوا haploid بإحدى مراحل حياتهم بعدين يصيروا diploid او العكس.. المهم انهم بقدروا ما يضلوا ثابتين

- Bacteria and Archaea
 - haploid only
 - Asexual reproduction by
 - Binary fission
 - Budding
 - Filamentous (like filamentous fungi)
 - all must replicate and segregate the genome prior to division

لازم يتم تضاعف الجينوم وانفصاله قبل ما يتم انقسام الخلية

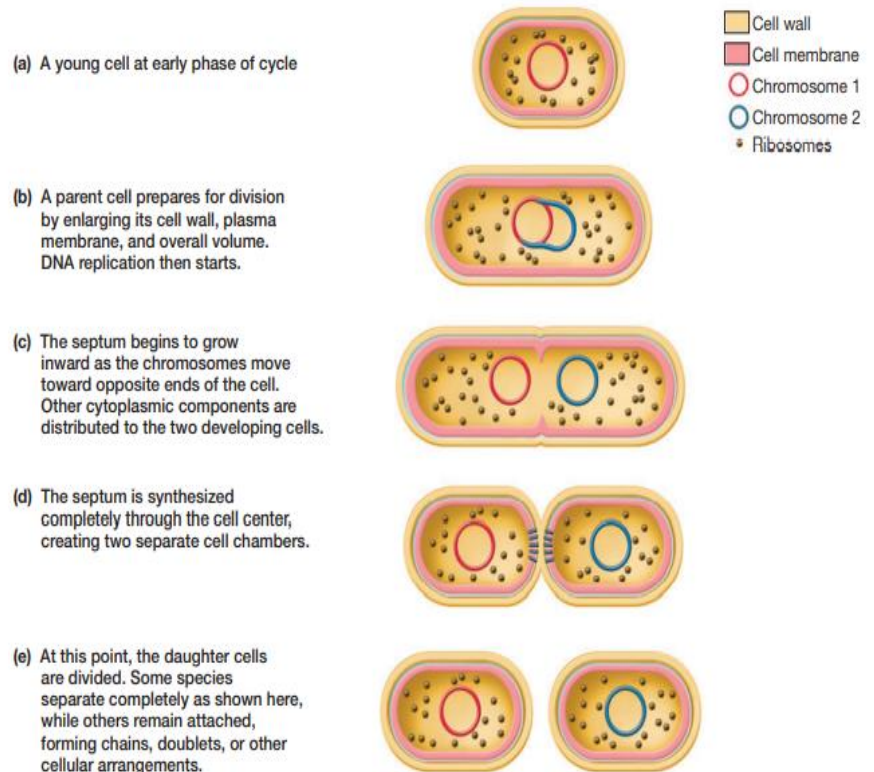


Figure 7.1 Binary Fission.

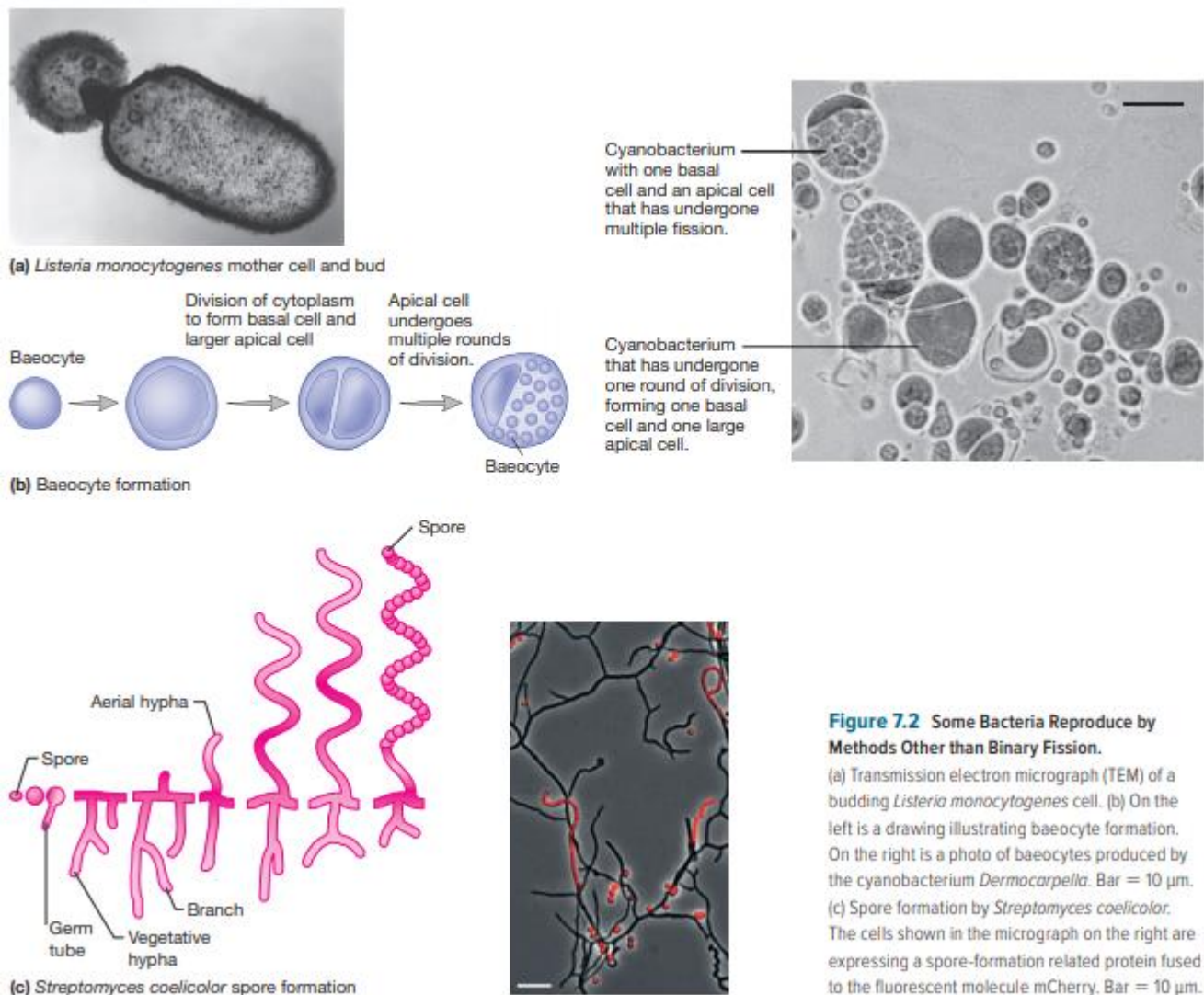


Figure 7.2 Some Bacteria Reproduce by Methods Other than Binary Fission.

(a) Transmission electron micrograph (TEM) of a budding *Listeria monocytogenes* cell. (b) On the left is a drawing illustrating baeocyte formation. On the right is a photo of baeocytes produced by the cyanobacterium *Dermocarpella*. Bar = 10 μm . (c) Spore formation by *Streptomyces coelicolor*. The cells shown in the micrograph on the right are expressing a spore-formation related protein fused to the fluorescent molecule mCherry. Bar = 10 μm .

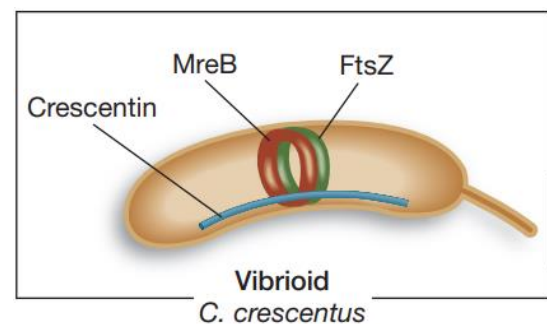
Slide 15:

Vibrio Cell Wall Growth and Cell Shape Determination

Vibrio (comma-shaped bacteria)

- FtsZ - forms Z ring
- MreB - helical polymerization throughout cell
- crescentin - intermediate filament homologue
- localizes to short, curved side of cell
- asymmetric cell wall synthesis forms curve

نفس الي انذكر بنشابت 3



🌸 Environmental factors that affect microbial growth

Growth: an increase in size or number

- Most organisms grow in fairly moderate environmental conditions

→ **Extremophiles**: bacteria that grow under harsh conditions that would kill most other organisms

معظم الكائنات بتعيش بظروف بيئية معتدلة ولكن في انواع من البكتيريا بتقدر تعيش في بيئات "قاسية" الكائنات الثانية ما بتقدر تعيش فيها.. هذول منسميهم **extremophiles** والههم أهمية صناعية كبيرة.

Table 7.2 Microbial Responses to Environmental Factors		
Descriptive Term	Definition	Representative Genera and Species
Solute Concentration and Water Activity		
Halophile	Requires high levels of sodium chloride, usually above about 0.2 M, to grow	<i>Halobacterium</i> , <i>Dunaliella</i> , <i>Ectothiorhodospira</i>
Osmotolerant	Able to grow over wide ranges of water activity or osmotic concentration	<i>Staphylococcus aureus</i> , <i>Saccharomyces rouxii</i>
Xerophile	Organisms that grow best at low water activity, typically with optima at 0.85 or below	<i>Xeromyces bisporus</i>
pH		
Acidophile	Growth optimum between pH 0 and 5.5	<i>Sulfolobus</i> , <i>Picrophilus</i> , <i>Ferroplasma</i> , <i>Acontium</i>
Neutrophile	Growth optimum between pH 5.5 and 8.0	<i>Escherichia</i> , <i>Euglena</i> , <i>Paramecium</i>
Alkaliphile	Growth optimum between pH 8.0 and 11.5	<i>Bacillus alcalophilus</i> , <i>Natronobacterium</i>
Temperature		
Psychrophile	Grows at 0°C and has an optimum growth temperature of 15°C or lower	<i>Bacillus psychrophilus</i> , <i>Chlamydomonas nivalis</i>
Psychrotolerant	Can grow at 0–7°C; has an optimum between 20 and 30°C and a maximum around 35°C	<i>Listeria monocytogenes</i> , <i>Pseudomonas fluorescens</i>
Mesophile	Has growth optimum between 20 and 45°C	<i>Escherichia coli</i> , <i>Trichomonas vaginalis</i>
Thermophile	Can grow at 55°C or higher; optimum often between 55 and 65°C	<i>Geobacillus stearothermophilus</i> , <i>Thermus aquaticus</i> , <i>Cyanidium caldarium</i> , <i>Chaetomium thermophile</i>
Hyperthermophile	Has an optimum between 85 and about 113°C	<i>Sulfolobus</i> , <i>Pyrococcus</i> , <i>Pyrodictium</i>
Oxygen Concentration		
Obligate aerobe	Completely dependent on atmospheric O ₂ for growth	<i>Micrococcus luteus</i> , most protists and fungi
Facultative anaerobe	Does not require O ₂ for growth but grows better in its presence	<i>Escherichia</i> , <i>Enterococcus</i> , <i>Saccharomyces cerevisiae</i>
Aerotolerant anaerobe	Grows equally well in presence or absence of O ₂	<i>Streptococcus pyogenes</i>
Obligate anaerobe	Does not tolerate O ₂ and dies in its presence	<i>Clostridium</i> , <i>Bacteroides</i> , <i>Methanobacterium</i>
Microaerophile	Requires O ₂ levels between 2–10% for growth and is damaged by atmospheric O ₂ levels (20%)	<i>Campylobacter</i> , <i>Spirillum volutans</i> , <i>Treponema pallidum</i>
Pressure		
Piezophile (barophile)	Growth more rapid at high hydrostatic pressures	<i>Photobacterium profundum</i> , <i>Shewanella benthica</i>

دونت ووري لا تحفظوه من هون.. رح نشرحه بالصفحات الي جاي / (^v^)(^v^)\

The most important factors that affect microbial growth:

- A. Solutes and water activity
- B. PH
- C. Temperature
- D. Oxygen level
- E. Pressure
- F. Radiation

A. Solutes and water activity*

*Water activity → how much water is available for microbes

- Changes in osmotic concentrations in the environment may affect microbial cells

- if a microorganism is placed in a Hypotonic solution (lower osmotic concentration):

→ water enters the cell

➡ Cell Dies!

→ cell swells and may burst

- if a microorganism is placed in a Hypertonic solution (higher osmotic concentration):

→ water leaves the cell

➡ Cell Dies!

→ membrane shrinkage from cell wall (plasmolysis) may occur

→ How do microbes adapt to changes in osmotic concentrations?

in hypotonic solutions → They reduce osmotic concentration of the cytoplasm by using **mechanosensitive (MS) channels** in plasma membrane that allow solutes to leave

خلينا نفكر فيها شوي .. هالأ عنا hypotonic solution يعني تركيز ال solute برا الخلية قليل والمي كتيرة وجوا الخلية تركيز ال solute عالي والمي قليلة .. بالوضع الطبيعي المي رح تدخل لجوا بفعل ال osmosis وتتفجر الخلية ف شو ممكن تعمل الخلية عشان ما يصير هيك (٦-٦) ؟

بتبش تطلع ال solute الي جوا عشان تخلي تركيزه قليل عن طريق قنوات منسجها **mechanosensitive (MS) channels** وهيك المي ما بتفوت بكميات كبيرة فبتحمي الخلية من الانفجار والموت

In hypertonic solutions → they **increase internal solute concentration** with compatible solutes to increase their internal osmotic concentration and these solutes are compatible with metabolism and growth

بال hypertonic solutions عنا العكس (برا الخلية مي قليلة و solute عالي وجوا الخلية تركيز ال solute قليل والمي كتيرة) فبتبش الخلية يا اما تصنع solute جوا الخلية كتير عشان تمنع المي من انها تطلع او بتاخده من برا وهيك ما بصير لها shrinkage. هدول ال solutes ما بأتروا بال growth وال metabolism تبعت البكتيريا

- Halophiles: they are an extremely adapted microbes that grow optimally in the presence of NaCl or other salts at a concentration above about 0.2M

- according to salt concentration, we have four groups of bacteria:

- Nonhalophiles: any concentration of salts would kill them
- Halotolerant: can survive with or without salt
- Moderate halophile
- Extreme halophiles:
 - they require salt concentrations of 2M and 6.2M
 - extremely high concentrations of potassium
 - cell wall, proteins, and plasma membrane require high salt to maintain stability and activity

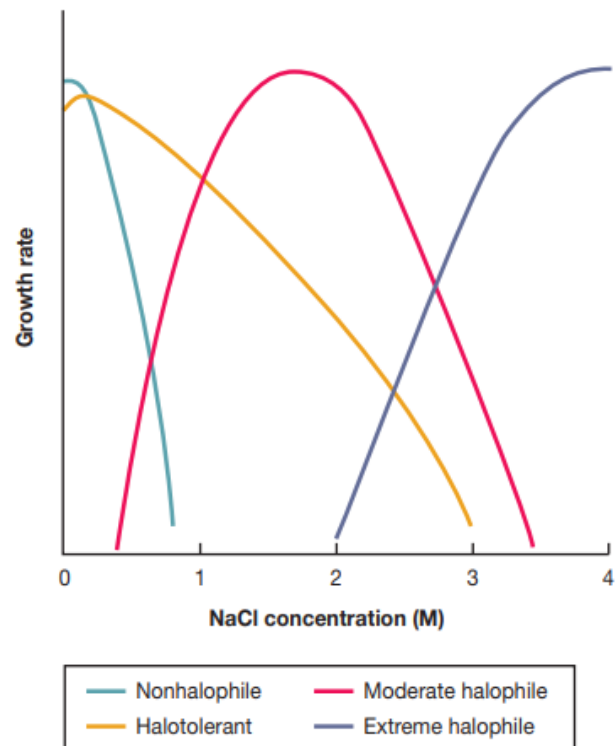


Figure 7.11 The Effects of Sodium Chloride on Microbial Growth. Four different patterns of microbial dependence on NaCl concentration are depicted. The curves are only illustrative and are not meant to provide precise shapes or salt concentrations required for growth.

- water activity:

When solutes such as salts and sugars are present, the water is "tied up" by its interaction with the solutes. Microbiologists express quantitatively the degree of water availability by determining water activity (aw).

لما يكون في solutes مثل الأملاح والسكريات ..المى يكون تفاعلها او توافرها محدود نتيجة تفاعلها وارتباطها مع هذول ال solutes لهنك ال microbiologists بعبروا عن درجة توافر المياه كميأ عن طريق تحديد ال water activity (aw) يعنى هاد المصطلح بعبر عن اديه في مي متاحة للمحيط الي موجودة فيه (او بالأحرى بحالتنا البكتيريا)

- The water activity of a solution is 1/100 the relative humidity of the solution (when expressed as a percent).
- It is also equivalent to the ratio of the solution's vapor pressure (P_{soln}) to that of pure water (P_{water})

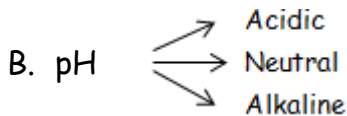
$$\rightarrow Aw = P_{\text{soln}} / P_{\text{water}}$$

Important points to consider:

- Low water activity means most water is bound
- Water activity is reduced by the interaction with solute molecules (osmotic effect)
 - higher [solute] $\Rightarrow$ lower aw
- reduced by adsorption to surfaces (matric effect)

➤ the highest $a_w = 1 \rightarrow$ Water without solutes

- **Osmotolerant microbes** : microbes that can grow over wide ranges of water activity



- It is the measure of the relative acidity of a solution and it equals the negative logarithm of the hydrogen ion concentration

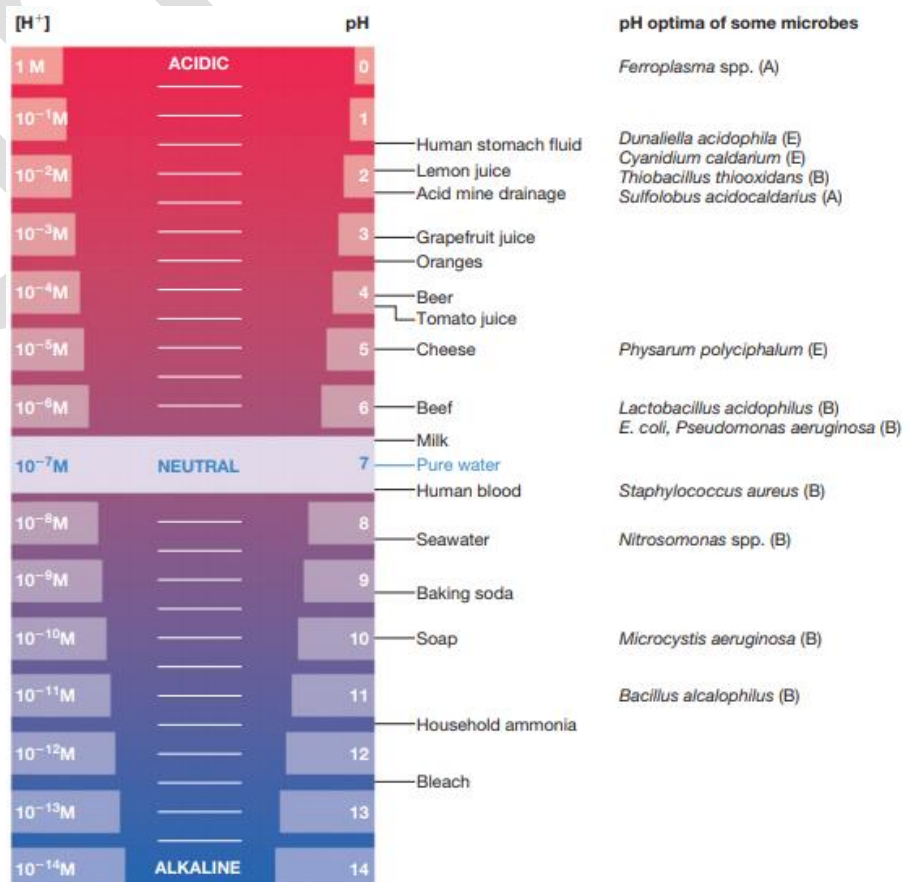
$$\text{pH} = -\log[\text{H}^+] = \log(1/[\text{H}^+])$$

- Each species has a definite pH growth range and pH growth optimum:
 - Acidophiles: growth optimum between pH 0 -5.5 (only one bacteria $\rightarrow$ *Helicobacter pylori*)
 - Neutrophiles: growth optimum between pH 5.5 - 7
 - Alkaliphiles (alkalophiles) : growth optimum between pH 8.5 -11.5 (alkaline solutions: any salt added to water $\rightarrow$ ex. *Vibrio*)

➔ Pathogenic bacteria are usually Neutrophiles

- pH Tolerance Mechanisms of Microbes

- Most microbes maintain an internal pH near neutrality
 - the plasma membrane is impermeable to proton
 - By exchanging potassium for protons
- Acidic tolerance response
 - pump protons out of the cell
 - some synthesize acid and heat shock proteins that protect proteins from these acidic conditions
- Many microorganisms change the pH of their habitat by producing acidic or basic waste products



C. Temperature

- Microbes cannot regulate their internal temperature so they are particularly susceptible to external temperatures
- Enzymes have optimal temperature at which they function optimally
- Organisms exhibit distinct cardinal growth temperatures
 - minimal
 - maximal
 - optimal

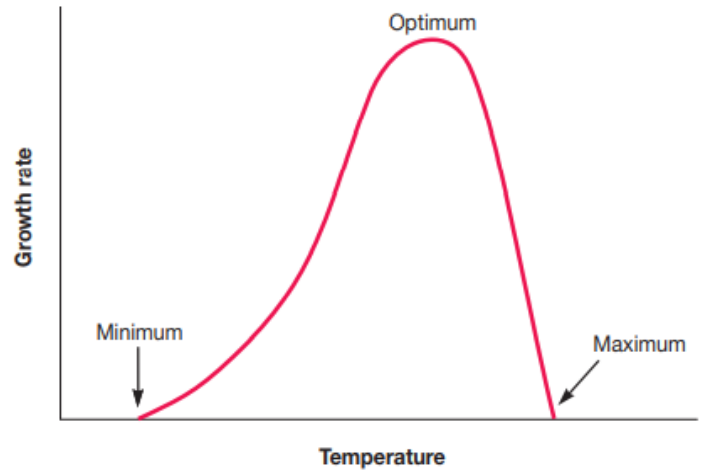


Figure 7.13 Temperature and Growth. The effect of temperature on growth rate.

كل انزيم الو درجة حرارة بشتغل فيها على الصورة الامثل منسيميا

ال optimum temperature. لما تقل درجة الحرارة عن

ال optimum بوقف الانزيم عن العمل وبتزيد سرعة تفاعله كل ما

قرب لل optimum. لما كل الانزيمات الي بالبكتيريا تزيد سرعة

تفاعلها، ال metabolism بشكل كلي بتصير more active ف بتتمو البكتيريا بشكل اسرع. ولما تزيد درجة الحرارة عن

ال optimum بصير النمو ابطء وازا زادت كثير تؤدي الى موت البكتيريا.

بسبب هاي التأثيرات..النمو تبع البكتيريا الو درجة حرارة مميزة منسيميا cardinal growth temperatures وفي منها

Minimum و maximal و optimal. (figure 7.13) (خدوا نظرة على table 7.3 بسلايد رقم 30)

- Temperature Ranges for Microbial Growth:

- Psychrophiles : 0° C to 20° C
- Psychrotrophs : 0° C to 35° C
- Mesophiles : 20° C to 45° C
- Thermophiles : 55° C to 85° C
- Hyperthermophiles : 85° C to 113° C

- Adaptations of Thermophiles:

- 1- A Protein structure stabilized by a variety of means:

- e.g. more H bonds
- e.g. more proline
- e.g. chaperones

- 2- they have Histone-like proteins that stabilize the DNA

- 3- a Membrane that is stabilized by variety of means:

- e.g. more saturated, more branched and higher molecular weight lipids
- e.g. ether linkages (in archaeal membranes)

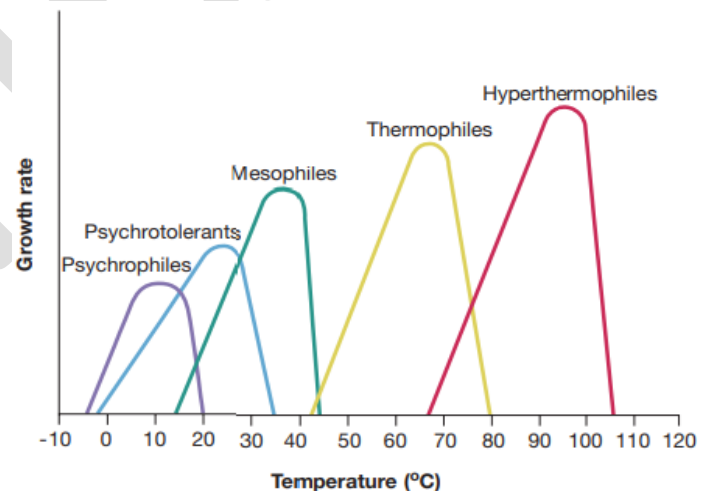
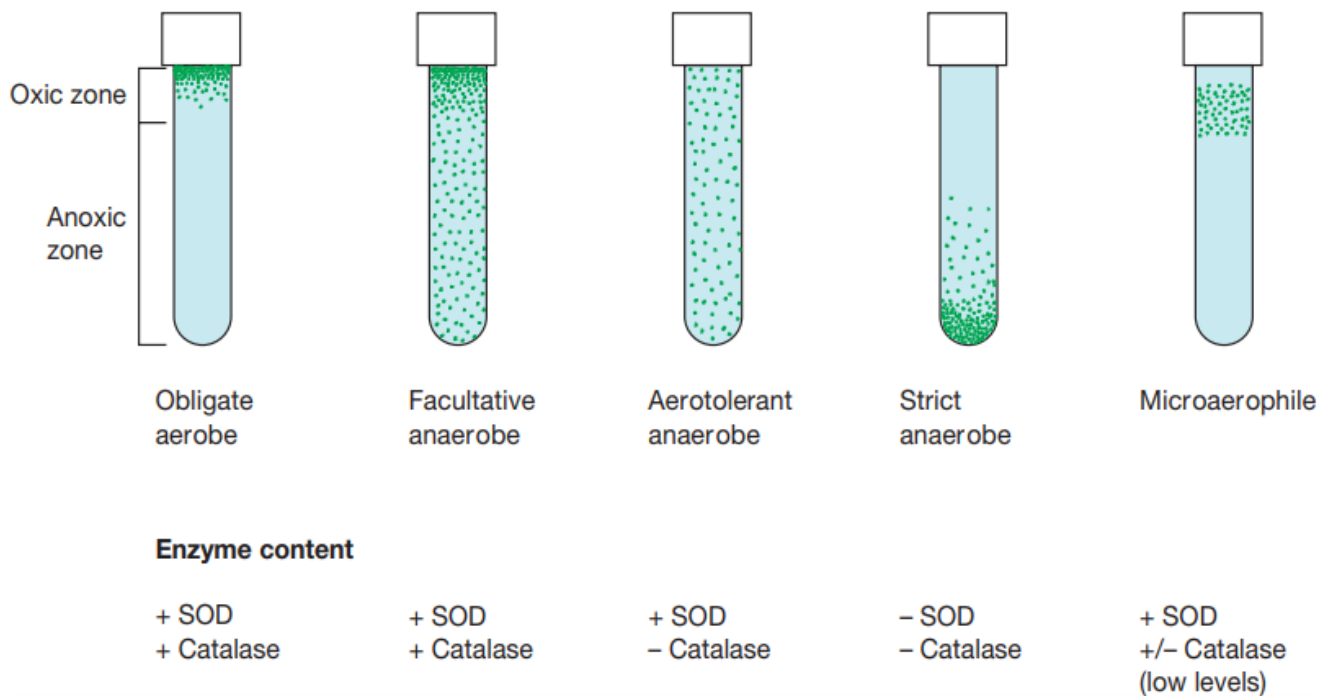


Figure 7.14 Temperature Ranges for Microbial Growth.

Microorganisms are placed in different classes based on their temperature ranges for growth. They are ranked in order of increasing growth temperature range as psychrophiles, psychrotolerants, mesophiles, thermophiles, and hyperthermophiles. Representative ranges and optima for these five types are illustrated.

D. Oxygen Concentration

- Growth in presence of different oxygen concentrations depends on a microbe's metabolic processes, electron transport chains (ETC), terminal electron acceptor used
- Aerobes: grows in presence of atmospheric oxygen (O_2) which is 20% O_2
- Anaerobes: grow in the absence of O_2
- Five types of relationships to oxygen have been described:
 - i. Obligate aerobes: require oxygen highly
 - ii. Obligate (strict) anaerobe: usually killed in presence of O_2 (O_2 is toxic for them)
 - iii. Facultative anaerobes: do not require O_2 but grow better in its Microaerophile: requires 2-10% O_2
 - iv. Aerotolerant anaerobes: grow with or without O_2



- The Basis of Different Oxygen Sensitivities:
 - Oxygen has unpaired electrons in the outer shell that make it inherently unstable and is usually reduced to reactive oxygen species (ROS):
 - superoxide radical O_2^-
 - hydrogen peroxide H_2O_2
 - hydroxyl radical $\cdot OH$
 - Aerobes produce enzymes that protect them from ROS (Protective enzymes):
 - superoxide dismutase (SOD)
 - catalase $\rightarrow$ breaks H_2O_2 into $H_2O + O_2$
 - peroxidase

- Strict anaerobic microbes lack/have very low quantities of
 - superoxide dismutase
 - Catalase

Because they can't tolerate O_2 in the first place

E. Pressure

- Microbes that live on land and water surface live at a pressure equals 1 atmosphere (atm)
- Even though, some Bacteria and Archaea live in deep sea with very high hydrostatic pressures
- Barotolerant: microbes adversely affected by increased pressure, but not as severely as nontolerant organisms
- Barophilic (piezophilic) organisms
 - They require or grow more rapidly in the presence of increased pressure
→ How? By changing membrane fatty acids to adapt to high pressures

F. Radiation

- Ionizing radiation : radiation of very short wavelength and high energy, which causes atoms to lose electrons (ionize)
- Two major forms: x-rays and gamma rays
- They produce mutations → death (sterilization)
- They cause a variety of changes in the cell disrupting the chemical structure of many molecules, including DNA

- damage may be repaired by DNA repair mechanisms if small dose

كميات قليلة من ال radiation بتسبب mutations ممكن الخلية انها تعملها تصليح (repair) عن طريق عمليتين:

- Light repair: breaks the bond between thymine dimers
- Dark repair: removes the damaged piece of DNA and replace it with a natural one

ولكن لما تزيد ال mutations هاد يؤدي الى قتل البكتيريا وهاي طريقة من طرق التعقيم (sterilization)

- *Deinococcus radiodurans* → extremely resistant to DNA damage

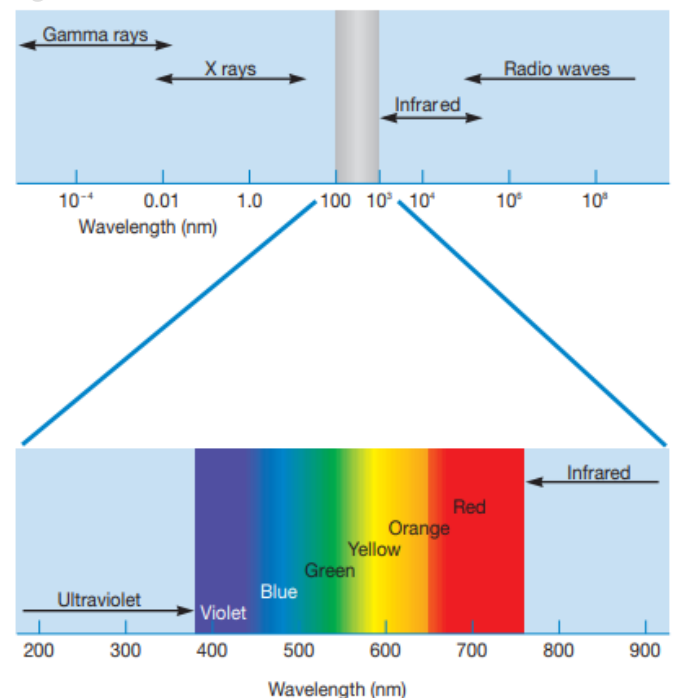


Figure 7.16 The Electromagnetic Spectrum. A portion of the spectrum showing the visible light range and the nearby ultraviolet and infrared wavelengths is expanded at the bottom of the figure.

- Ultraviolet (UV) radiation
 - The most lethal UV radiation has a wavelength of 260 nm, the wavelength most effectively absorbed by and damaging to DNA
 - mutations → death
 - causes formation of **thymine dimers** in DNA
 - requires direct exposure on microbial surface
 - DNA damage can be repaired by several repair mechanisms
- Visible light
 - at high intensities generates **singlet oxygen** (1O_2) → a powerful oxidizing agent
 - many microbes that live on light-exposed surfaces use **carotenoid** pigments to be protected from photooxidation

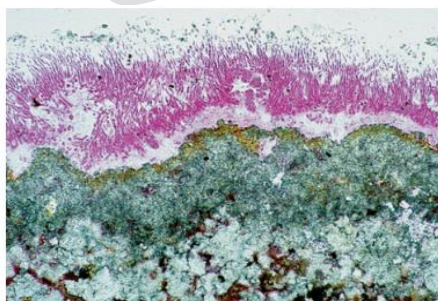
🌿 Microbial Growth in natural environments and in the lab

Microbial Growth in Natural Environments

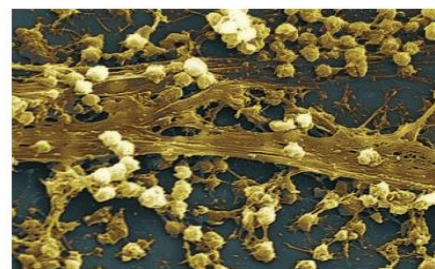
- Microbial environments are complex, constantly changing, often contain low nutrient concentrations (oligotrophic environment) and may expose a microorganism to overlapping gradients of nutrients and environmental factors
- Most microbes grow attached to surfaces (sessile) rather than free floating (planktonic)
 - These attached microbes are members of complex, slime enclosed communities called a

Biofilm

- made up of proteins, polysaccharides and DNA
 - when all these are secreted by the bacteria, they are called Extracellular Polymeric Substance (EPS)
- made by bacteria to protect itself from environmental conditions
- Biofilms are ubiquitous (found everywhere) in nature in water
- Can be formed on any conditioned surface



(a) Biofilm on surface of a stromatolite



(b) Biofilm on internal surface of an indwelling catheter

Figure 7.17 Examples of Biofilms. Biofilms form on almost any surface exposed to microorganisms. (a) Biofilm on the surface of a stromatolite in Walker Lake (Nevada, USA), an alkaline lake. The biofilm consists primarily of the cyanobacterium *Calothrix* sp. (b) *Staphylococcus aureus* biofilm on the luminal surface of an indwelling catheter; colorized SEM X2,363.

Laboratory Culture of Cellular Microbes

- **Culture Media:** is a solid or liquid preparation used to grow, transport, and store microorganisms

→ to be effective, the medium must contain all the nutrients the microorganism requires for growth.

- Culture media can be classified based on several parameters:

1- the chemical constituents

from which they are made

according to chemical constituents we have 2 types of media:

- Defined (synthetic) media
- Complex media

2- their physical nature

we have 3 types of media:

- liquid
- semisolid
- supportive

3- their function

4 types:

- Supportive (general purpose)
- Enriched
- Selective
- Differential

1- a. Defined (synthetic) media

you know exactly what this media contains and what is the concentration of these constituents

Table 7.4 Types of Media	
Basis for Classification	Types
Chemical composition	Defined (synthetic), complex
Physical nature	Liquid, semisolid, solid
Function	Supportive (general purpose), enriched, selective, differential

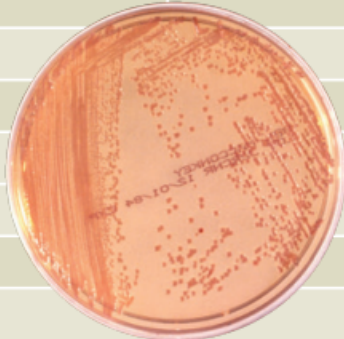
Table 7.5 Examples of Defined Media			
BG-11 Medium for Cyanobacteria	Amount (g/liter)	Medium for <i>Escherichia coli</i>	Amount (g/liter)
NaNO ₃	1.5	Glucose	1.0
K ₂ HPO ₄ · 3H ₂ O	0.04	Na ₂ HPO ₄	16.4
MgSO ₄ · 7H ₂ O	0.075	KH ₂ PO ₄	1.5
CaCl ₂ · 2H ₂ O	0.036	(NH ₄) ₂ SO ₄	2.0
Citric acid	0.006	MgSO ₄ · 7H ₂ O	200.0 mg
Ferric ammonium citrate	0.006	CaCl ₂	10.0 mg
EDTA (Na ₂ Mg salt)	0.001	FeSO ₄ · 7H ₂ O	0.5 mg
Na ₂ CO ₃	0.02	Final pH 6.8–7.0	
Trace metal solution ¹	1.0 ml/liter		
Final pH 7.4			

b. Complex Media

- contain undefined components such as Peptones → protein hydrolysates from protein sources and Extracts → aqueous extracts like meat extract, and yeast extract

→ you can't tell exactly what this media contains and what is the concentration of each component
→ used when you don't know exactly what a certain type of bacteria needs or when you grow fastidious bacteria

- Three commonly used complex media are nutrient broth, tryptic soy broth, and MacConkey

Table 7.6 Some Common Complex Media							
Nutrient Broth	Amount (g/liter)		Tryptic Soy Broth	Amount (g/liter)		MacConkey Agar	Amount (g/liter)
Peptone (gelatin hydrolysate)	5		Tryptone (pancreatic digest of casein)	17		Pancreatic digest of gelatin	17.0
Beef extract	3		Peptone (soybean digest)	3		Pancreatic digest of casein	1.5
Final pH 6.8			Glucose	2.5		Peptic digest of animal tissue	1.5
			Sodium chloride	5		Lactose	10.0
			Dipotassium phosphate	2.5		Bile salts	1.5
			Final pH 7.3			Sodium chloride	5.0
						Neutral red	0.03
						Crystal violet	0.001
						Agar	13.5
MacConkey Agar						Final pH 7.4	

agar

2- Both liquid and solidified media are routinely used in laboratories

Agar: - the most commonly used solidifying agent

- sulfated polysaccharide solidifying agent
- used to convert liquid media to the solid form
- extracted from red seaweed
- it is an excellent hardening agent because most microorganisms cannot degrade it.

Liquid culture is called
broth culture



3- Functional Types of Media

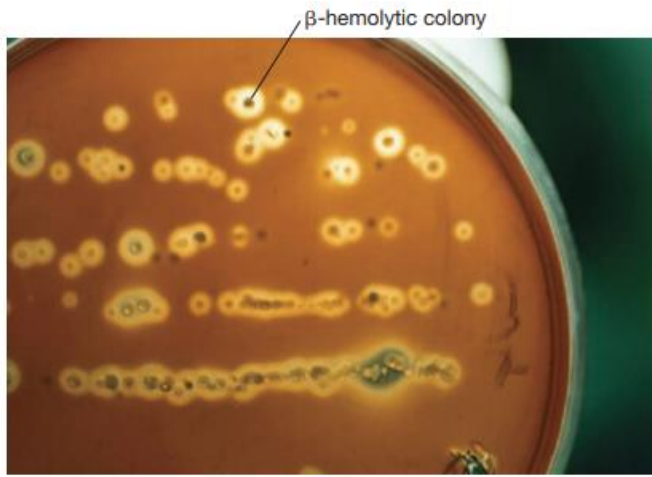
a. Supportive or general purpose media (e.g. tryptic soy agar(TSA), tryptic soy broth & Nutrient agar):

Any bacteria can live on it; it supports the growth of many microorganisms

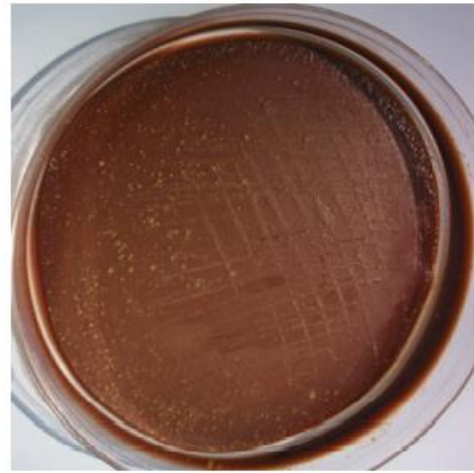
b. Enriched media:

Supportive media+ Blood → Enriched media (e.g blood agar)

It encourages the growth of pathogenic and fastidious microbes



(a) Blood agar



(b) Chocolate agar

Figure 7.22 Enriched Media. (a) Blood agar culture of bacteria from the human throat. (b) Chocolate agar is used to grow fastidious organisms such as *Neisseria gonorrhoeae*. The brown color is the result of heating red blood cells and lysing them before adding them to the medium. It is called chocolate agar because of its chocolate brown color.

c. Selective media:

Favor the growth of some microorganisms and inhibit growth of others

e.g : MacConkey agar → selects for Gram-negative bacteria (suppress the growth of Gram-positive bacteria) because it contains bile salts and crystal violet that are toxic for Gram+ve

d. Differential Media

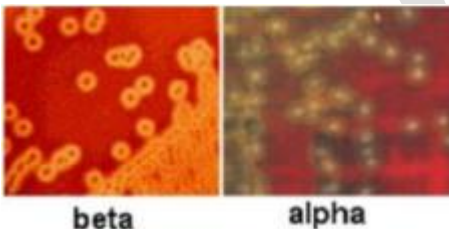
Distinguish between different groups of microorganisms based on their biological characteristics

- e.g. blood agar
 - hemolytic versus nonhemolytic bacteria

في بعض انواع من البكتيريا بتعمل complete hemolysis لخلايا الدم؛ بسمي هاي العملية **β-hemolysis**. ومنمیزها عن طريق تكون clear zone حول الكولونيز

وفي بعض انواع بتعمل incomplete hemolysis لخلايا الدم وهاد منمسيه **α-hemolysis** ومنشوفو على شكل greenish color حول الكولونيز تبعت البكتيريا

بناءا على ال blood hemolysis بقدر امیز بين انواع مختلفة من البكتيريا حسب نوع ال hemolysis الي هي بتعملو لهيك اسمها differential.



- e.g. MacConkey agar



- lactose fermenters versus nonfermenters

ازا البكتيريا بتعمل fermentation للاكتوز بتغير لون الميديا من اصفر لاحمر عند الكولونيز

Table 7.7 Mechanisms of Action of Selective and Differential Media

Medium	Functional Type	Mechanism of Action
Blood agar	Enriched and differential	Blood agar supports the growth of many fastidious bacteria. These can be differentiated based on their ability to produce hemolysins—proteins that lyse red blood cells. Hemolysis appears as a clear zone (β -hemolysis) or greenish halo around the colony (α -hemolysis) (e.g., <i>Streptococcus pyogenes</i> , a β -hemolytic streptococcus).
Eosin methylene blue (EMB) agar	Selective and differential	Two dyes, eosin Y and methylene blue, inhibit the growth of Gram-positive bacteria. They also react with acidic products released by certain Gram-negative bacteria when they use lactose or sucrose as carbon and energy sources. Colonies of Gram-negative bacteria that produce large amounts of acidic products have a green, metallic sheen (e.g., fecal bacteria such as <i>E. coli</i>).
MacConkey (MAC) agar	Selective and differential	The selective components in MAC are bile salts and crystal violet, which inhibit the growth of Gram-positive bacteria. The presence of lactose and neutral red, a pH indicator, allows the differentiation of Gram-negative bacteria based on the products released when they use lactose as a carbon and energy source. The colonies of those that release acidic products are red (e.g., <i>E. coli</i>).
Mannitol salt agar	Selective and differential	A concentration of 7.5% NaCl selects for the growth of staphylococci. Pathogenic staphylococci can be differentiated based on the release of acidic products when they use mannitol as a carbon and energy source. The acidic products cause a pH indicator (phenol red) in the medium to turn yellow (e.g., <i>Staphylococcus aureus</i>).

انتبهوا انو ممكن النوع الواحد من ال media ينتمي لأكثر من تصنيف؛ زي ال blood agar هو enriched وبنفس الوقت differential

- Strict Anaerobic Microbes
- All strict anaerobic microorganisms lack or have very low quantities of superoxide dismutase & Catalase enzymes so they can't tolerate O_2
- Based on that, Anaerobes must be grown without O_2
→ So we grow them in Anerobic jars or an Anerobic incubator

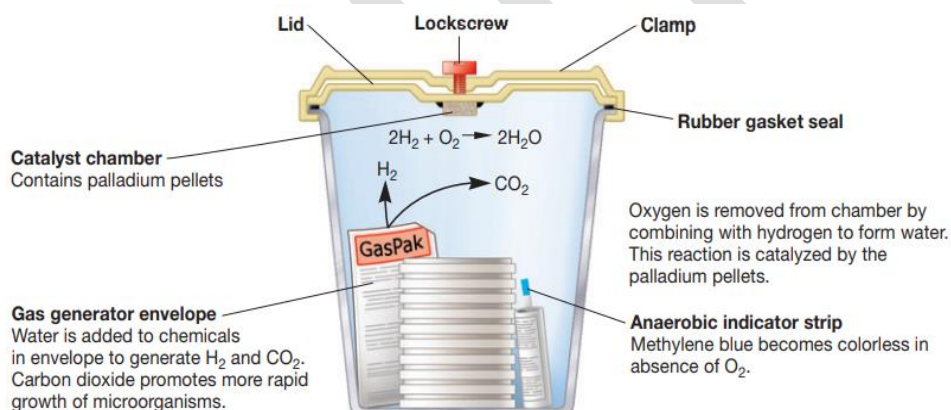


Figure 7.24 The GasPak Anaerobic System. Hydrogen and carbon dioxide are generated by a GasPak envelope. The palladium catalyst in the chamber lid catalyzes the formation of water from hydrogen and oxygen, thereby removing oxygen from the sealed chamber.

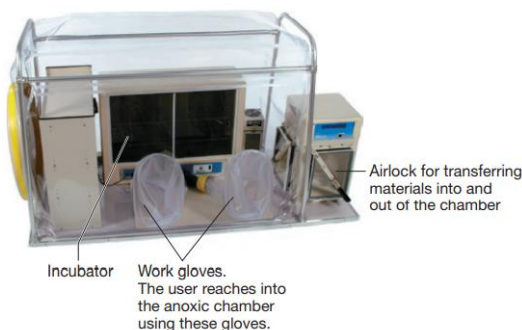


Figure 7.23 An Anaerobic Workstation and Incubator. This system

وا يجيبوا كل ال plates وبحطوهم ب jar وعلى الغطا تبعو بحطوا فازلين
عشان يمنعوا الهوا من انو يفوت. وعشان يتخلصوا من الاكسجين الي جوا كانوا
يحطوا شمعة جوا ال jar قبل ما يسكروه. لما يخلص الاكسجين الشمعة رح
تطفئ.. وهيك حيبطل في اكسجين نتيجة لتفاعل الاحتراق الي صار داخل الوعاء.
هلا بدل الشمعة بستخدموا Gas pak فيه مواد كيميائية منضيف عليها مي
بتعطينا هيدروجين و CO_2 .. ال H_2 بتفاعل مع الاكسجين الي جوا وبعطينا H_2O
وهيك يبطل في اكسجين داخل ال jar بالاضافة الي انهم استبدلوا الفازلين بغطاء
محكم الاغلاق زي ما هو مبين ب figure 7.24

• Isolation of Pure Cultures

عشان نقدر ندرس البكتيريا لازم نعرف كيف منقدر نعزلها عن بيئتها الاصلية ونجيبها للمختبر ونجهزها ليهيك لازم ندرس الطرق الي بتم فيها عزل pure cultures من البكتيريا

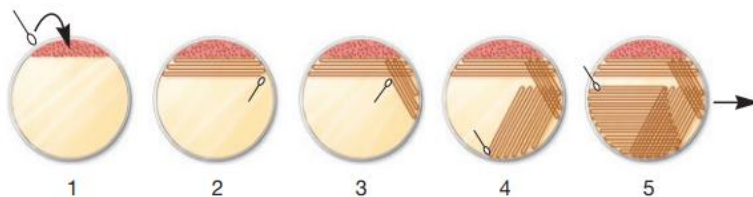
- Pure culture is a population of cells arising from a single cell developed by Robert Koch
- Allows for the study of single type of microorganism in mixed culture
- We have three techniques:
 - Streak plate
 - Spread plate
 - Pour plate

■ Streak plate:

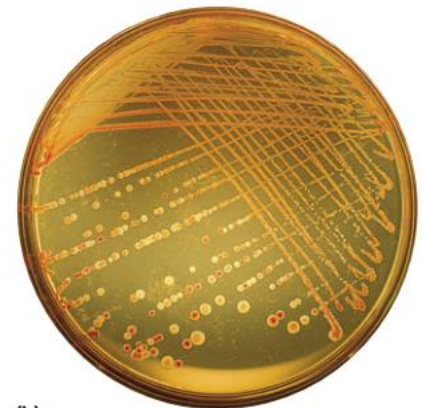
- Involves spreading a mixture of cells on an agar surface so that individual cells are well separated from each other using a bacteriological loop
- Each cell can reproduce to form a separate **colony** → visible growth or cluster of microorganisms
 - In this technique, cells are transferred to the edge of an agar plate with an inoculating loop or swab and then streaked across the surface in one of several patterns (figure 7.25).
 - After the first sector is streaked, the inoculating loop is sterilized and an inoculum for the second sector is obtained from the first sector.
 - A similar process is followed for streaking the third sector, except that the inoculum is from the second sector.
 - Thus this is essentially a dilution process. Eventually very few cells will be on the loop, and single cells will drop from it as it moves across the agar surface. These develop into separate colonies

اول اشي مناخذ عينة من البكتيريا م ال plate الاصلي باستخدام ال loop ومنعمل streaking (زي نفردھا) على اول جزء من plate جديد بعددين منعقم ال loop ومنحطھا على طرف اول منطقة عملناھا ومنعمل streaking على المنطقة الثانية منعيد الخطوة هاي عيل ما نوصل للنقطة 5 ال موضحة بالشكل. لما نعمل streaking نكون زي كانا عم نعمل تخفيف تدريجي كل مرة منعقم فيها ال loop فبتصير تبين معنا colonies منفردة وواضحة (please watch: https://www.youtube.com/watch?v=EqMz_jcd3us)

Note: This method only works if the spreading tool (usually an inoculating loop) is resterilized after each of steps 1–4.



(a) Steps in a streak plate



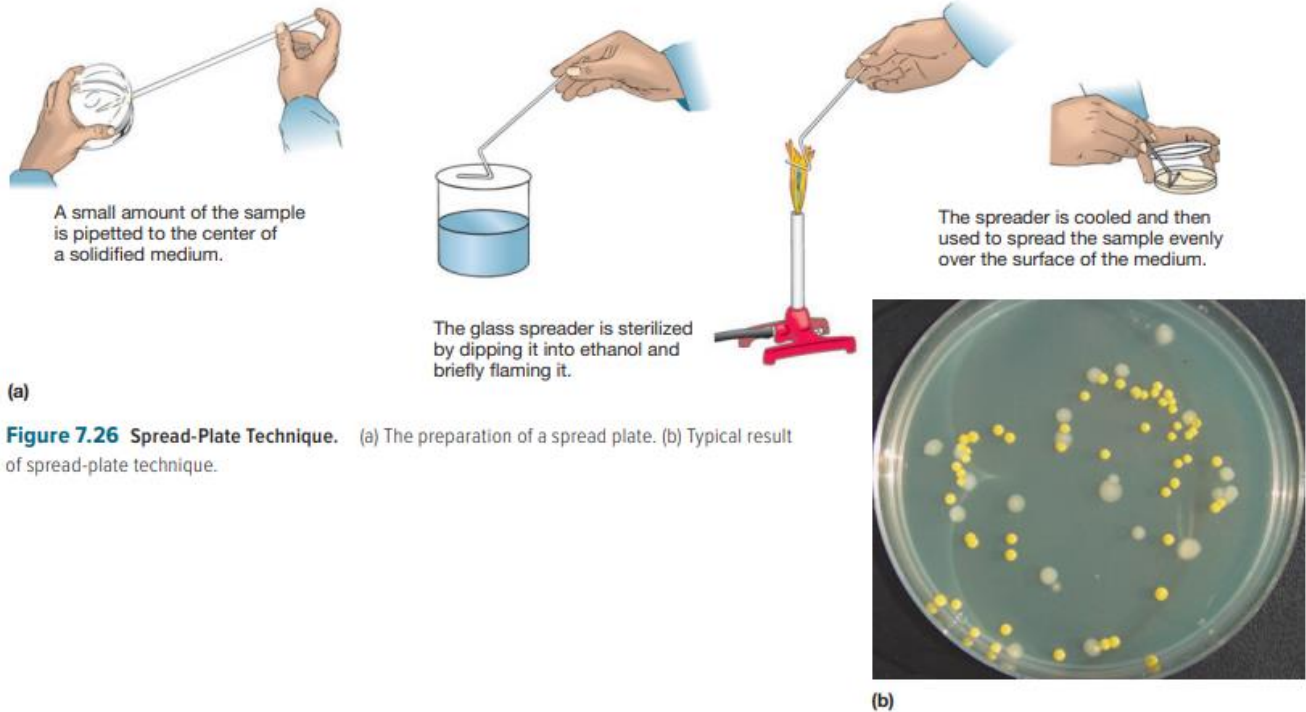
(b)

Figure 7.25 Streak-Plate Technique. A typical streaking pattern is shown (a), as well as an example of a streak plate (b).

■ Spread Plate

- small volume of diluted mixture containing approximately 30-300 cells is transferred to the center of an agar plate and spread evenly over the surface with a sterile L bent rod

يكون عنا بكتيريا ب Nutrient broth (نفس الـ nutrient agar بس بدون agar يعني سائلة) مناخذ منها شوي بال pipette ومنحطها بنص الـ plate الجديد. بعدين منجيب الـ L bent rod ومنعقمه ومنفرد فيه البكتيريا على كل الـ plate. بعد 24 ساعة بال incubator ملاحظ نمو البكتيريا على سطح الـ plate



(a)

Figure 7.26 Spread-Plate Technique. (a) The preparation of a spread plate. (b) Typical result of spread-plate technique.

■ Pour plate:

- the original sample is serially diluted to reduce the microbial population sufficiently to obtain separate colonies when plating
- Then small volumes of several diluted samples are mixed with liquid agar that has been cooled to about 45°C , and the mixtures are poured immediately into sterile culture dishes.

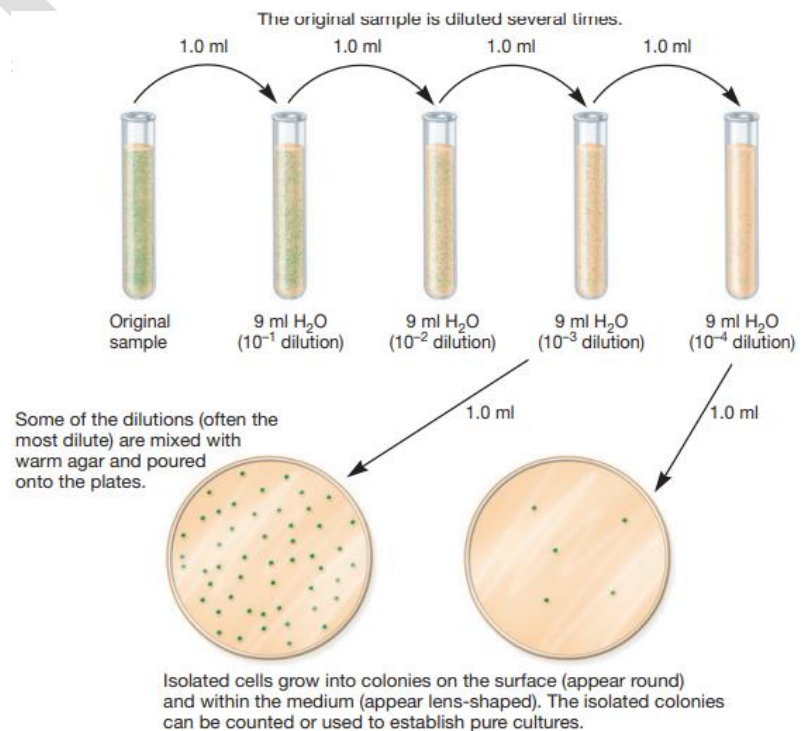


Figure 7.27 Pour-Plate Technique.

→ both Spread and pour plate may be used to determine the number of viable microorganisms in an original sample

🌸 Growth curves

Growth phases

- Growth: increase in cellular constituents that may result in:
 - increase in cell number
 - increase in cell size
- Growth refers to population growth rather than growth of individual cells
- How did they know those phases?

We take the bacteria to a batch culture

where we grow them without adding any

nutrients or removing wastes (grown in a closed system). they are left there for 2 or 3 days according to the type of bacteria. Every hour during these days, a sample of bacteria was taken and counted to see how many are there.

the logarithm of the number of viable cells versus the incubation time is plotted in a growth curve as in (figure 7.29)

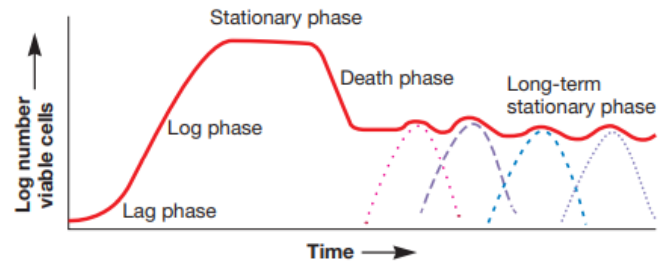


Figure 7.29 Microbial Growth Curve in a Closed System. The five phases of the growth curve are identified. The dotted lines shown during the long-term stationary phase represent successive waves of genetic variants that evolve during this phase of the growth curve.

- The Growth curve:
- Has five distinct phases:
 - 1st phase: **Lag phase** → Adaptation phase
 - The Cell is synthesizing new components
 - e.g. to replenish spent materials
 - e.g. to adapt to new medium or other conditions
 - ما يكون في اي نمو او انقسام للخلايا..بس بتكون عم تتكيف مع البيئة الجديدة
 - Varies in length
 - in some cases can be very short or even absent because we took the bacteria from the same kind of media and the same temperature as the new media

2nd phase: **Log phase** → Exponential phase

- Rate of growth and division is constant and maximal
- Population is most uniform in terms of chemical and physical properties during this phase
- During log phase, cells grow as quickly as they can for the conditions available
 - nutrient concentrations may be limiting, limiting growth
 - abundance of nutrients allows transport mechanisms to be saturated, so growth increase is minimal for adding in even more nutrients

لما يبيلش الغذاء يقل نتيجة استهلاك البكتيريا لـ... بتبلش البكتيريا تفوت بال **stationary phase** (عدد البكتيريا الميتة = عدد البكتيريا العايشة) ولكنهم وجدوا انو العكس غير صحيح (يعني اذا بزيد الغذاء ما بزيد النمو) لأنو بتكون ال **transport mechanisms** مشبعة وقد ما زدناها غذاء مستحيل تاخده

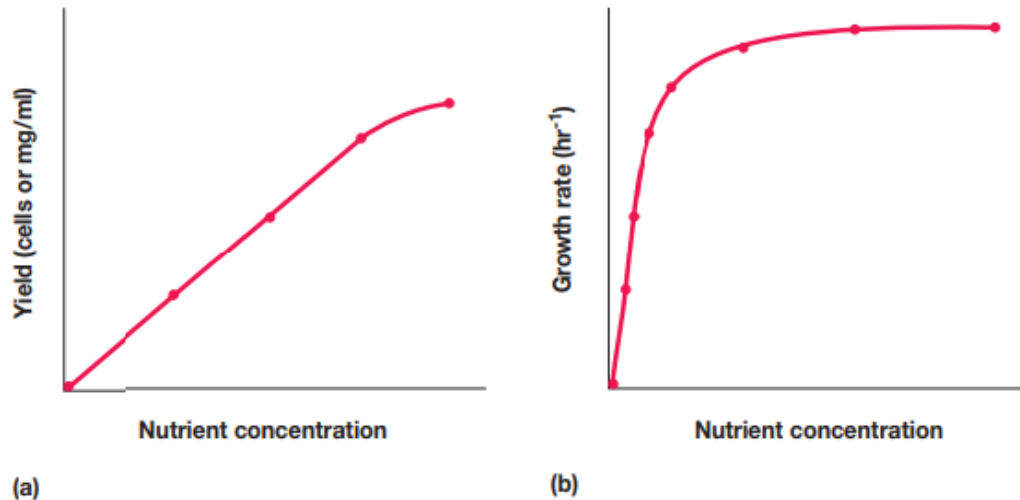


Figure 7.30 Nutrient Concentration and Growth. (a) The effect of changes in limiting nutrient concentration on total microbial yield. At sufficiently high concentrations, total growth will plateau. (b) The effect on growth rate.

3rd phase: **Stationary phase**

- Closed system population growth eventually ceases, total number of viable cells remains constant
 - active cells stop reproducing or reproductive rate is balanced by death rate
- Possible reasons for stationary phase
 - nutrient limitation
 - toxic waste accumulation
 - limited oxygen availability
 - critical population density reached

4th phase: **Death phase**

- We have two alternative hypothesis for this phase:

1- **cells are Viable But Not Culturable (VBNC)**

they found that sometimes cells are alive but dormant meaning that they are capable of new growth when conditions are right and suitable

2- **Programmed cell death**

They found that some cells commit suicide for other cells to grow using them as a nutrient (they are genetically programmed to die) (o_o) !

5th phase: **Prolonged Decline in Growth**

- Bacterial population continually evolves :
During this time, the bacterial population continually evolves so that actively reproducing cells are those best able to use the nutrients released by their dying brethren and best able to tolerate the accumulated toxins.
- Process marked by successive waves of genetically distinct variants
- Natural selection occurs

→ Those phases are not only seen in the lab..they can also occur in the environment.
(remember..bacteria lives in nutrient-poor environments)

يعني ممكن الغذاء يتوافر للبكتيريا بظرف معين وتمر بالlog phase بعدين يرجع بقل الغذاء فيتمر بالstationary phase وهكذا

• The Mathematics of Growth

▪ Generation (doubling) time

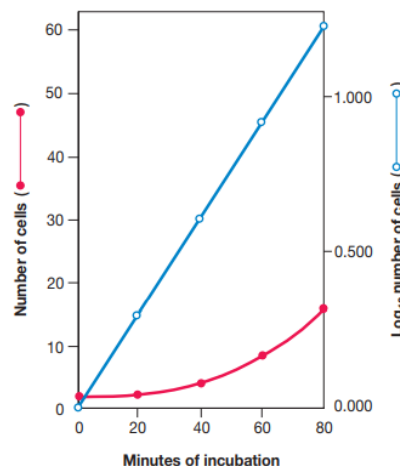
- time required for the population to double in size (divide by binary fission)
- varies depending on species of microorganism and environmental conditions
- range from 10 minutes for some bacteria to days for some eukaryotes

Number of cells at (t)= number of cells at (0) $\times 2^n$ = generation time

→ Log for both sides: **$\text{Log}_{(t)} = \log N_{(0)} + n \log 2$**

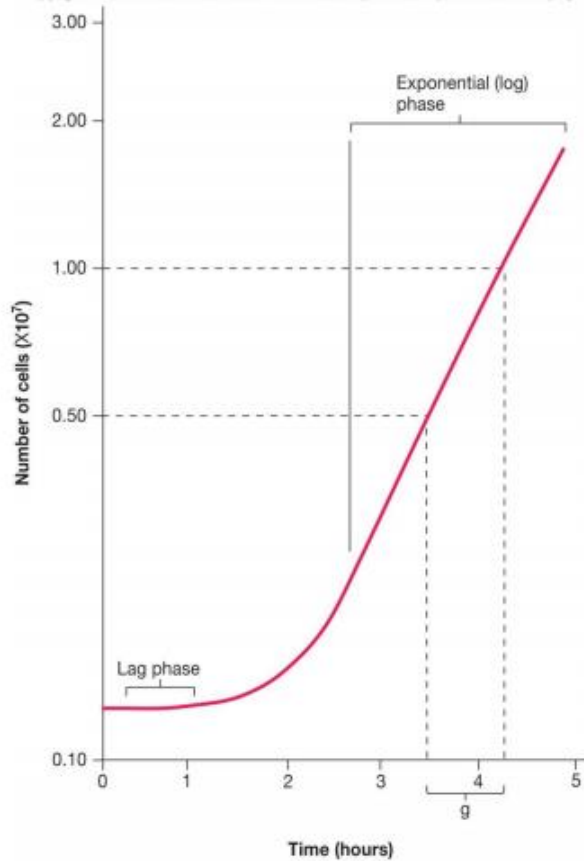
Table 7.8 An Example of Exponential Growth				
Time ¹	Division Number	2 ⁿ	Population ² (N ₀ × 2 ⁿ)	log ₁₀ N _t
0	0	2 ⁰ = 1	1	0.000
20	1	2 ¹ = 2	2	0.301
40	2	2 ² = 4	4	0.602
60	3	2 ³ = 8	8	0.903
80	4	2 ⁴ = 16	16	1.204

¹ The hypothetical culture begins with one cell having a 20-minute generation time.
² Number of cells in the culture.



Outcomes of Gen. Time Calculations

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Table 7.9 Examples of Generation Times¹

Microorganism	Incubation Temperature (°C)	Generation Time (Hours)
Bacteria		
<i>Escherichia coli</i>	40	0.35
<i>Bacillus subtilis</i>	40	0.43
<i>Staphylococcus aureus</i>	37	0.47
<i>Pseudomonas aeruginosa</i>	37	0.58
<i>Clostridium botulinum</i>	37	0.58
<i>Mycobacterium tuberculosis</i>	37	~12
<i>Treponema pallidum</i>	37	33
Protists		
<i>Tetrahymena gelii</i>	24	2.2–4.2
<i>Chlorella pyrenoidosa</i>	25	7.75
<i>Paramecium caudatum</i>	26	10.4
<i>Euglena gracilis</i>	25	10.9
<i>Giardia lamblia</i>	37	18
<i>Ceratium tripos</i>	20	82.8
Fungi		
<i>Saccharomyces cerevisiae</i>	30	2
<i>Monilia fructicola</i>	25	30

¹Generation times differ depending on the growth medium and environmental conditions used.

74

المهم تعرفوه من سلايد 74 انو ال generation time ممكن يكون بالساعات او بالأيام حسب نوع البكتيريا

❁ Measurement of Microbial Growth and Direct Measurement of Cell Numbers

Measurement of Microbial Growth:

- Can measure changes in number of cells in a population
- Can measure changes in mass of population

• Direct Measurement of Cell Numbers

- a) counting chambers
- b) on membrane filters
- c) electronic counters - flow cytometry

a) Counting Chambers

Counting chambers consist of specially designed slides and coverslips; the space between the slide and coverslip creates a chamber of known depth.

- Easy, inexpensive, and quick
- Useful for counting both eukaryotes and prokaryotes
- Cannot distinguish living from dead cells

Figure 7.34

كل مربع بعد كم خلية فيه

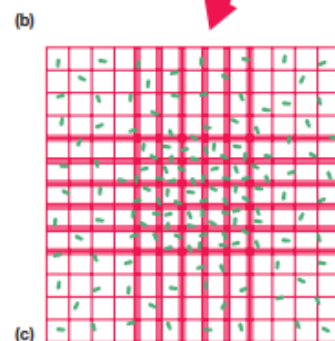
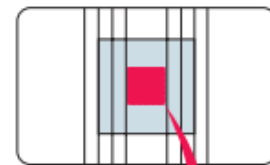
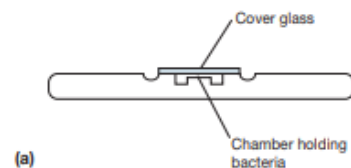


Figure 7.34 The Petroff-Hausser Counting Chamber.

b) Direct Counts on Membrane Filters

- Cells filtered through special membrane that provides dark background for observing cells
- Cells are stained with fluorescent dyes
- Useful for counting bacteria
- With certain dyes, can distinguish living from dead cells

c) Flow Cytometry

- Microbial suspension forced through small orifice with a laser light beam
- And then the movement of a microbe through orifice impacts electric current that flows through orifice. Suddenly, instances of disruption of current are counted
- Specific antibodies can be used to determine size and internal complexity
- It is the same process they used to count red blood cells

• Viable counting: Alive or dead?

It's hard to determine if a cell is alive or not; it isn't always a clear cut in microbiology because a cell can exist in a variety of states between 'fully viable' and 'actually dead'

- Viable counting methods are:
 1. Spread/Pour Plates
 2. Membrane Filter Method
 3. MPN Method

Alive

Actively metabolizing cell

Cell with reduced metabolic activity

Some metabolic activity and plasma membrane intact, but RNA content is reduced

Plasma membrane intact, but no detectable metabolic activity

Extensive damage to plasma membrane

Cellular DNA degraded

Cell fragments

Dead

1. Spread/Pour Plates

- diluted sample of bacteria is spread over solid agar surface or mixed with agar and poured into Petri plate
- after incubation the numbers of organisms are determined by counting the number of colonies multiplied by the dilution factor
- results expressed as colony forming units (CFU)

2. Membrane filter technique

- bacteria from aquatic samples are trapped on membranes
- then, the membrane is soaked in culture media
- as a result, colonies grow (meaning that they are viable) on membrane
- colony count determines number of bacteria in sample

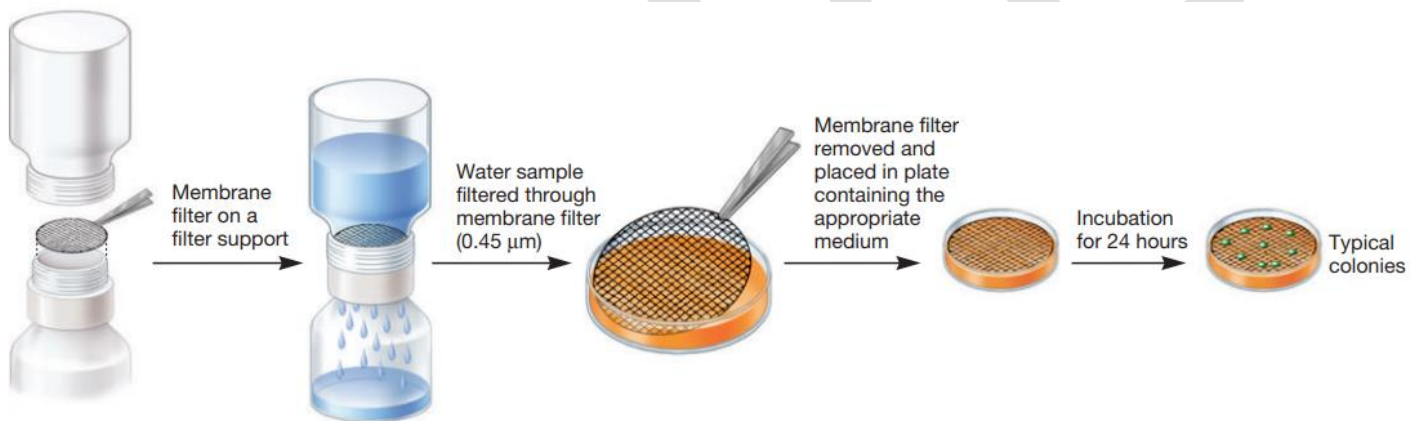


Figure 7.36 The Membrane Filtration Procedure. Membranes with different pore sizes are used to trap different microorganisms. Incubation times for membranes also vary with the medium and microorganism.

3. MPN (most probable number) Method

- If a microbe cannot be cultured on plate (solid) media
- Several dilutions of a culture are made and added to tubes containing a suitable liquid growth medium
- Turbidity determined to yield the most probable number (MPN)

• Measurement of Cell Mass

Can be done using three different ways:

- i. Dry weight → Cells growing in liquid medium are collected by centrifugation, washed, dried in an oven, and weighed.

→ time consuming

- ii. Quantity of a particular cell constituent
 → like : protein, DNA, ATP, or chlorophyll
 → useful if amount of substance in each cell is constant
- iii. Turbidometric measures (spectrophotometry)
 → depends on the fact that microbial cells scatter light that strikes them
 → it is quick, easy, and sensitive

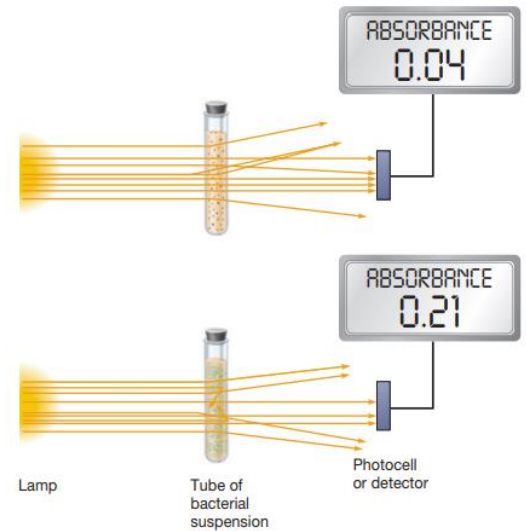


Figure 7.38 Turbidity and Microbial Mass Measurement. Determination of microbial mass by measurement of light absorption. As the population and turbidity increase, more light is scattered and the absorbance reading given by the spectrophotometer increases.

🌸 Chemostats and Turbidostats (Continuous Culture)

- Continuous culture
 Growth in an open system
 - continual provision of nutrients
 - continual removal of wastes

عكس ال batch تماماً، عبارة عن نظام مفتوح يتم إضافة الغذاء وإزالة الفضلات بشكل مستمر.

→ Maintains cells in log phase at a constant biomass concentration for extended periods

- Achieved using a continuous culture system:
 - They maintain a microbial populations in exponential (log) phase , growing at a known rate and at a constant biomass concentration for extended periods
 - Continuous culture systems make possible the study of microbial growth at very low nutrient levels, concentrations close to those present in natural environments.
 - These systems make it possible to study the interactions of microbes under conditions resembling those in aquatic environments
 - Continuous culture systems also are used in food and industrial microbiology
- Two types of continuous culture systems:
 - A. Chemostats
 - B. Turbidostat

A. Chemostats:

- Rate of incoming medium = rate of removal of medium from vessel
→ The culture medium for a chemostat has a limited quantity of an essential nutrient (e.g., a vitamin). Because one nutrient is limiting, growth rate is determined by the rate at which sterile medium is fed into the growth chamber; the final cell density depends on the concentration of the limiting nutrient.

يحتوي الـ culture media الـ chemostat على كمية محدودة من nutrients أساسية ولأنه في واحد من هذول الـ nutrients محدود فيسببه يتم تحديد أو تحجيم الـ growth rate . كنتيجة .. كثافة الخلايا النهائية بتعتمد على تركيز هذول الـ nutrients .

Dilution Rate and Microbial Growth

Dilution rate: the rate at which medium flows through the culture vessel relative to the vessel volume figure 7.40

- cell density maintained at wide range of dilution rates
- Chemostat operates best at low dilution rate

B. Turbidostat

- Regulates flow rate of media through vessel to maintain predetermined turbidity or cell density

من الـ Turbidity بقدر اعرف كثافة الخلايا الموجودة بالـ media (يعني عددها بالنسبة لحجم الميديا)

- Differs from chemostat in:

→ dilution rate varies where as in chemostat is constant

→ NO limiting nutrient

→ Turbidostat operates best at high dilution rate; a chemostat is most stable and effective at lower dilution rates.

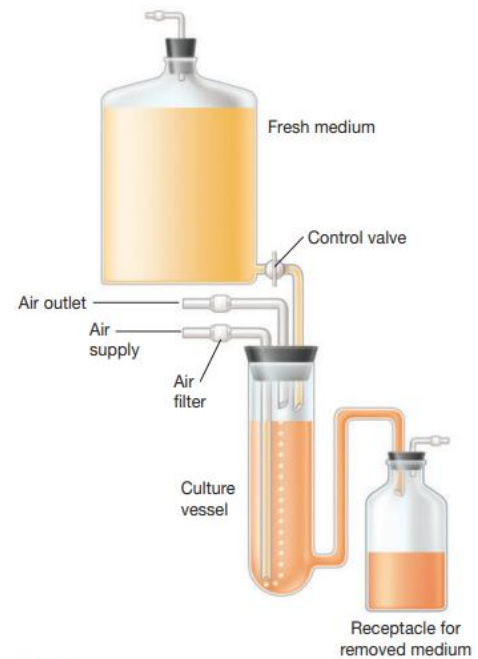


Figure 7.39 A Chemostat. Schematic diagram of the system. The fresh medium contains a limiting amount of an essential nutrient. Although not shown, usually the incoming medium is mechanically mixed with that already in the culture vessel. Growth rate is determined by the flow rate of medium through the culture vessel.

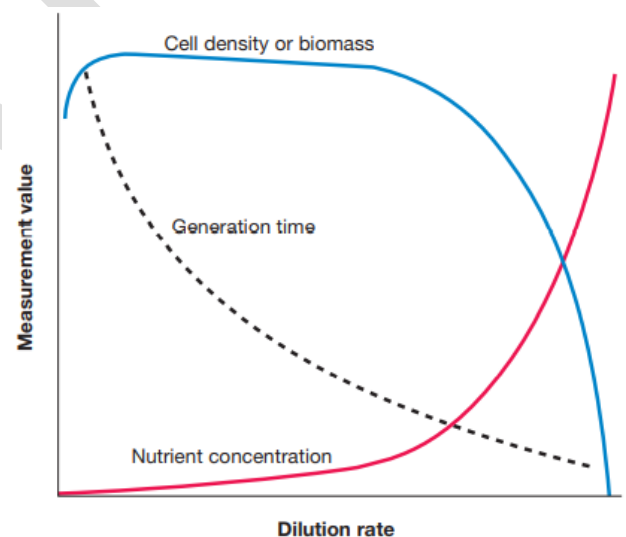


Figure 7.40 Chemostat Dilution Rate and Microbial Growth. The effects of changing the dilution rate in a chemostat.

The End

(~)♡(~)